

R. Kasaija & Partners — Programmer Guide

Project: Law Firm Web Application (Full Stack)
Version: 1.0.0
Stack: React 19 · Vite 8 · Tailwind CSS 4 · Express · MySQL · Google Gemini AI
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1. Project Overview

This is the official website and internal admin system for R. Kasaija & Partners Advocates, a Kampala-based law firm.

What the application does

Layer	Purpose
Public site	Marketing pages, practice areas, team profiles, appointment booking, contact form, AI legal intake chat

Admin panel	Login, dashboard, appointments, clients, cases, tasks, billing, AI FAQ/memory, reports
API	Express REST endpoints + SSE streaming for AI chat
Database	MySQL (kasaija_lawfirm) for all persistent data

Tech stack summary

Frontend: React + Vite + React Router + Framer Motion + Tailwind CSS 4 + Lucide icons

Backend: Node.js + Express + express-session + bcrypt + mysql2

AI: Google Generative AI (Gemini)

Hosting: Vercel (frontend + serverless API) or XAMPP (local)

2. Directory Structure

LAW FIRM WEB 2/

```
├── api/           # Node.js Express backend (primary)
|   ├── index.js   # App entry, session, route mounting
|   ├── _db.js     # MySQL connection pool
|   └── routes/
|       ├── contact.js    # POST /api/contact
|       ├── appointments.js # POST /api/book
|       ├── admin.js      # /api/admin/* (CMS)
|       └── chat.js       # POST /api/chat/stream (SSE)
|
├── src/           # React frontend source
|   ├── main.jsx     # React root mount
|   ├── App.jsx      # Router, layout, lazy routes
|   ├── index.css     # Tailwind theme + global styles
|   ├── App.css       # (minimal / legacy)
|   ├── pages/        # Route-level pages
|   |   ├── Home.jsx
|   |   ├── About.jsx
|   |   ├── Practice.jsx
|   |   └── Team.jsx
```

- | | └─ Book.jsx
- | | └─ Contact.jsx
- | | └─ Admin.jsx
- | └─ components/ # Reusable UI blocks
- | | └─ Navbar.jsx
- | | └─ Footer.jsx
- | | └─ Hero.jsx
- | | └─ Marquee.jsx
- | | └─ GVM.jsx
- | | └─ PracticeGrid.jsx
- | | └─ WhyUs.jsx
- | | └─ ChatWidget.jsx
- | | └─ CustomCursor.jsx
- | └─ data/
- | | └─ constants.js # Practice areas, staff, clients
- | └─ utils/
- | └─ api.js # apiFetch + streamFetch helpers
- |
- └─ public/assets/img/ # Firm photos & logo (served statically)
- └─ dist/ # Production build output (gitignored ideally)
- └─ php_backup/ # Legacy PHP admin + db_init
- └─ api_php_backup/ # Legacy PHP API endpoints
- └─ server/ # Alternate CommonJS server (legacy duplicate)
- └─ index.html # Vite HTML shell + Google Fonts
- └─ vite.config.js # Base path: /RKASAIJA/
- └─ tailwind.config.js # Extended theme tokens
- └─ vercel.json # SPA + API rewrites
- └─ package.json # Scripts & dependencies

3. Setup & Installation

Prerequisites

- Node.js 18+ and npm
- XAMPP (or standalone MySQL) for local database
- Google Gemini API key (for AI chat)

Step-by-step local setup

Step 1 — Clone / open project

```
cd "c:\xampp\htdocs\LAW FIRM WEB 2"
```

Step 2 — Install dependencies

```
npm install
```

Step 3 — Initialize the database

Start MySQL in XAMPP, then open in browser:

```
http://localhost/LAW%20FIRM%20WEB%202/php_backup/db_init.php
```

This creates database kasaija_lawfirm and all tables. Default admin:

Field	Value
Username	admin
Password	kasaija@2026

Step 4 — Create .env in project root

```
DB_HOST=localhost
```

```
DB_USER=root
```

```
DB_PASSWORD=
```

```
DB_NAME=kasaija_lawfirm
```

```
PORT=5000
```

```
SESSION_SECRET=your_random_secret_here
```

```
FRONTEND_URL=http://localhost:5173
```

```
GEMINI_API_KEY=your_gemini_api_key
```

```
GEMINI_MODEL=gemini-1.5-flash
```

```
VITE_API_URL=http://localhost:5000/api
```

Step 5 — Run backend

```
npm run server
```

Server listens on <http://localhost:5000>.

Step 6 — Run frontend

npm run dev

Frontend runs on http://localhost:5173 (Vite default).

Step 7 — Production build

npm run build

npm run preview

Output goes to dist/.

4. Environment Variables

Variable	Used by	Purpose
DB_HOST	API	MySQL host
DB_USER	API	MySQL user
DB_PASSWORD	API	MySQL password
DB_NAME	API	Database name (default: kasaija_lawfirm)
PORT	API	Server port (default: 5000)
SESSION_SECRET	API	Express session signing key
FRONTEND_URL	API	CORS allowed origin
NODE_ENV	API	production enables secure cookies
GEMINI_API_KEY	API	Google AI key
GEMINI_MODEL	API	Model name (default: gemini-1.5-flash)
VITE_API_URL	Frontend	API base URL (build-time)

5. Design System — Colors, Typography & Layout

The site uses a warm legal boutique aesthetic: cream backgrounds, dark brown text, gold accents.

5.1 Color palette

Defined in two places (keep them aligned when editing):

src/index.css — @theme block (Tailwind v4 primary source):

Token	Hex	Usage
cream	#FDFBF7	Page background
dark	#2A1D10	Primary text, buttons, hero overlays
gold	#B8956A	Accents, highlights

gold-mid	#9E7F5A	Section labels, italic accent text
light	#F6EDDA	Section gradients, cards

tailwind.config.js — extended tokens:

Token	Hex	Notes
darker	#1F1308	Footer background
light2	#E8D8B4	Success cards, gradients
border	rgba(139,111,71,0.2)	Subtle dividers
shadow	rgba(61,40,23,0.15)	Soft shadows

Hero exception: Hero section uses near-black #060301 with cinematic overlays, not cream.

5.2 Typography

Loaded in index.html:

Role	Font	Tailwind class
Headlines	Fraunces (serif)	font-serif
Body / UI	Inter Tight (sans)	font-sans (default)
Labels / codes	JetBrains Mono	font-mono

Patterns:

- Section labels: font-mono text-[10px] tracking-widest uppercase text-gold-mid — often prefixed with §
- Headlines: font-serif text-[clamp(...)] leading-[0.95] tracking-tight
- Emphasis: .accent → italic gold-mid (<em className="accent">)

5.3 Layout & spacing

Container (max width + horizontal padding):

.container-custom → max-w-[1400px] mx-auto px-6 lg:px-12

Page top padding: Most pages use pt-24 lg:pt-32 to clear the fixed navbar.

Navbar: Fixed top, z-50, becomes more opaque on scroll (bg-cream/95 backdrop-blur-xl).

Grid patterns:

- Footer: lg:grid-cols-12
- Admin stat cards: responsive flex/grid
- Practice grid: responsive card grid via PracticeGrid.jsx

5.4 Buttons

Class	Style
.btn-primary	Dark gradient pill, cream text, lift on hover
.btn-outline	Border-only pill, context-colored

5.5 Animations

Source	Animation
index.css	beamSway, pDrift, scrollLine, ringPulse, letterUp
tailwind.config.js	marquee, fade-up, fade-in
Framer Motion	Page sections: whileInView, initial/animate
Hero	Ken Burns scale on team photo, particle drift, light beams

5.6 Custom cursor

CustomCursor.jsx + .cursor-dot in CSS — desktop only (hidden lg:block), grows gold on hover targets.

5.7 Background texture

Body has subtle SVG fractal noise overlay at 2.2% opacity for paper-like feel.

6. Frontend Architecture

6.1 Entry flow

index.html → main.jsx → App.jsx

6.2 Routing (App.jsx)

Path	Page	Navbar/Footer
/	Home	Yes
/about	About	Yes
/practice	Practice	Yes
/team	Team	Yes
/book	Book	Yes
/contact	Contact	Yes
/admin	Admin	No (standalone layout)

Features:

- Lazy loading — all pages loaded via React.lazy()
- ScrollToTop — scrolls to top on route change
- ChatWidget — hidden on /admin

- CustomEvent open-chat — Navbar/Hero trigger AI chat globally

6.3 API client (src/utils/api.js)

- apiFetch(endpoint, options) — JSON REST calls
- streamFetch(endpoint, body, onChunk, onDone, onError) — SSE parser for AI chat

Base URL: `import.meta.env.VITE_API_URL || '/api'`

6.4 Asset paths

Vite base is `/RKASAIJA/` — all public assets use:

`/RKASAIJA/assets/img/firm_logo.jpeg`

Staff photos in `constants.js` follow the same prefix.

Images live in `public/assets/img/`.

7. Page-by-Page Guide

7.1 Home (src/pages/Home.jsx)

Sections (top to bottom):

1. Hero — cinematic full-screen intro
2. Marquee — scrolling Latin legal maxims
3. GVM — mission/values block
4. Practice preview — first 6 practice areas via `PracticeGrid limit={6}`
5. WhyUs — differentiators
6. Trusted counsel — client name list from `CLIENTS`
7. CTA — book consultation / open chat

Key interactions:

- `navigate('/practice')` for practice areas
 - `window.dispatchEvent(new CustomEvent('open-chat'))` for AI
-

7.2 About (src/pages/About.jsx)

Firm history, values, and team overview. Uses same design tokens: cream bg, serif headlines, motion on scroll.

7.3 Practice (src/pages/Practice.jsx)

Data source: `PRACTICE_AREAS` from `constants.js`

Behavior:

- Grid of 12 practice areas via PracticeGrid
- Click area → expands detail panel (AnimatePresence)
- Deep link support: `navigate('/practice', { state: { selectedId: 'banking' } })`
- Auto-scrolls to #area-detail-view

Each area has: id, num, name, desc, lawyer, long.

7.4 Team (src/pages/Team.jsx)

Data source: STAFF array

Each member: name, role, focus, bio, creds[], photo.

Layout: card grid with photo, credentials list, hover animations.

7.5 Book (src/pages/Book.jsx)

Form fields → API mapping:

Form field	API field
name	client_name
email	client_email
phone	client_phone
area	practice_area
lawyer	preferred_lawyer
date	preferred_date
time	preferred_time
message	message

Endpoint: POST /api/book

Success state: Thank-you card with gradient from-[#F6EDDA] to-[#E8D8B4].

7.6 Contact (src/pages/Contact.jsx)

Endpoint: POST /api/contact

Required: name, email, message

Optional: phone, practice area

7.7 Admin (src/pages/Admin.jsx)

Standalone login + dashboard. No public navbar/footer.

Tabs: Overview, Appointments, (Clients/Cases/Settings stubs in UI)

Uses apiFetch('/admin/login'), /admin/overview, /admin/appointments.

8. Shared Components

Component	File	Role
Navbar	Navbar.jsx	Fixed nav, mobile menu, active link underline, AI button
Footer	Footer.jsx	Dark footer, visit/contact/links columns
Hero	Hero.jsx	Full-screen hero, beams, particles, animated headline
Marquee	Marquee.jsx	Infinite horizontal scroll of legal phrases
GVM	GVM.jsx	Mission / vision / motto section
PracticeGrid	PracticeGrid.jsx	Reusable practice area cards (limit, onSelect, selectedId)
WhyUs	WhyUs.jsx	Firm differentiators
ChatWidget	ChatWidget.jsx	Floating AI chat, SSE streaming
CustomCursor	CustomCursor.jsx	Desktop custom cursor

ChatWidget integration pattern

// Open from anywhere:

```
window.dispatchEvent(new CustomEvent('open-chat'));
```

Listens in ChatWidget.jsx via useEffect + window.addEventListener('open-chat', ...).

9. Static Data & Content

File: src/data/constants.js

Edit practice areas

Add/modify objects in PRACTICE_AREAS:

```
{  
  id: "unique-slug",  
  num: "13",  
  name: "Display Name",  
  desc: "Short card description",  
  lawyer: "Lead Lawyer Name",  
}
```

long: "Full detail paragraph for expanded view"

}

Edit team members

Modify STAFF array — update photo path to match /RKASAIJA/assets/img/counsel_*.jpeg.

Edit client marquee

Modify CLIENTS string array on Home page.

10. Backend API

Entry: api/index.js

Session configuration

- Cookie name: rkp_session
- Store: MySQL via express-mysql-session
- Duration: 24 hours
- Admin session shape: { user, name, role }

Public endpoints

Method	Path	Description
GET	/api/health	Health check
POST	/api/contact	Contact form submission
POST	/api/book	Appointment booking
POST	/api/chat/stream	AI chat (SSE)

Admin endpoints (require session)

Method	Path	Description
POST	/api/admin/login	Admin login
POST	/api/admin/logout	Destroy session
GET	/api/admin/overview	Dashboard stats
GET	/api/admin/appointments	List appointments
POST	/api/admin/appointments/update	Update status/notes
GET	/api/admin/clients	List/search clients
GET	/api/admin/clients/:id	Client detail + cases
POST	/api/admin/clients	Create client

GET	/api/admin/cases	List/filter cases
GET	/api/admin/cases/:id	Case detail + notes + tasks
POST	/api/admin/cases	Open new case
GET	/api/admin/tasks	Task list
GET	/api/admin/billing	Invoices
GET	/api/admin/billing/stats	Billing totals
GET	/api/admin/faq	FAQ cache
POST	/api/admin/faq/toggle	Enable/disable FAQ
GET/POST	/api/admin/memory	AI knowledge base
GET	/api/admin/reports	Analytics
POST	/api/admin/settings/change-password	Password change

Row-Level Security (RLS)

Non-super_admin users only see records where assigned_lawyer matches their username or display name. Implemented in applyRLS() in admin.js.

ID sequencing

- Clients: CLT-YYYY-001
- Cases: KP-YYYY-001

11. Database Schema

Database: kasaija_lawfirm

Init script: php_backup/db_init.php

Tables overview

Table	Purpose
appointments	Online booking requests
contact_submissions	Contact form entries
activity_log	Audit trail (public + admin actions)
admin_users	Admin accounts (bcrypt passwords)
login_attempts_log	Login rate limiting support
clients	Client registry
cases	Case management
case_notes	Case timeline notes

tasks	Tasks & deadlines
billing	Invoices
faq_cache	Cached AI FAQ responses
ai_memory	Admin-managed AI knowledge
token_usage	Daily AI token tracking

Key enums

appointments.status: pending, confirmed, cancelled, completed

cases.status: open, in_progress, awaiting_docs, hearing_scheduled, settlement, closed, won, lost

cases.priority: low, medium, high, urgent

billing.status: draft, sent, paid, overdue, cancelled

12. Admin Dashboard

Login flow

1. User submits username/password
2. POST /api/admin/login validates against admin_users.password_hash (bcrypt)
3. Session stored; frontend sets isLoggedIn = true
4. Dashboard fetches /api/admin/overview

Dashboard widgets

- Appointment totals (pending, confirmed, unread)
- Contact enquiry totals
- Active case count
- Recent activity log (15 entries)

Changing admin password

POST /api/admin/settings/change-password with current + new + confirm (min 8 chars).

13. AI Chat System

Architecture

ChatWidget (React)

→ streamFetch('/chat/stream')

→ api/routes/chat.js

→ Google Generative AI (Gemini)

→ SSE chunks back to browser

System prompt

Large embedded SYSTEM_PROMPT in chat.js includes:

- Firm details, hours, contact
- All 12 practice areas with lead lawyers
- All 7 advocate profiles
- Ugandan legal context & disclaimers
- Booking guidance & emergency escalation

SSE format

```
data: {"type":"content","content":"partial text"}
```

```
data: {"type":"done"}
```

```
data: {"type":"error","message":"..."}
```

Frontend streaming

streamFetch in api.js reads the response body stream, parses data: lines, and calls callbacks.

14. Deployment

Vite base path

vite.config.js:

```
base: '/RKASAIJA/'
```

Deploy under subdirectory /RKASAIJA/ on your host, or change base to / for root deployment.

Vercel

vercel.json rewrites:

- /api/* → api/index.js (serverless)
- /* → index.html (SPA)

Set all env vars in Vercel dashboard.

XAMPP static hosting

1. npm run build
 2. Copy dist/ contents to web root (e.g. htdocs/RKASAIJA/)
 3. Run API separately or proxy to Node server
-

15. Legacy / Backup Code

Folder	Description
php_backup/	Original PHP admin + db_init.php
api_php_backup/	PHP API equivalents
server/	CommonJS duplicate of Express API

Use api/ (ES modules) as the canonical backend. PHP backups are reference only.

16. Common Tasks (Step-by-Step)

Add a new page

1. Create src/pages/NewPage.jsx
2. Lazy-import in App.jsx: `const NewPage = lazy(() => import('./pages/NewPage'));`
3. Add route: `<Route path="/new" element={<NewPage />} />`
4. Add link in Navbar.jsx navLinks array
5. Match styling: pt-24 lg:pt-32 pb-24 bg-cream min-h-screen + container-custom

Add a team member

1. Add photo to public/assets/img/
2. Add entry to STAFF in constants.js
3. Optionally update AI system prompt in api/routes/chat.js

Change brand colors

1. Update @theme in src/index.css
2. Update tailwind.config.js theme.extend.colors
3. Search for hardcoded hex values (Hero #060301, gradients) and update if needed

Add API endpoint

1. Create or extend route file in api/routes/
2. Import and mount in api/index.js
3. Call from frontend via `apiFetch('/your-endpoint')`

17. Troubleshooting

Issue	Likely cause	Fix
Images 404	Wrong base path	Ensure paths use <code>/RKASAIJA/assets/img/...</code>

API CORS errors	FRONTEND_URL mismatch	Set FRONTEND_URL to exact dev URL
Admin login fails	DB not initialized	Run db_init.php
Chat not responding	Missing GEMINI_API_KEY	Add key to .env, restart server
Session not persisting	Cookie/credentials	Use credentials: 'include' if needed; check same-site
Build blank page	Wrong base	Match deploy path to vite.config.js base